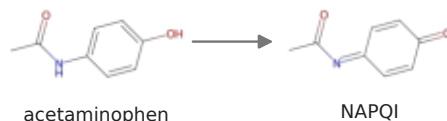
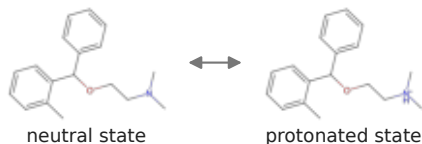


A

Chemical-state and metabolic expansion

state enumeration

metabolic transformation



Nodes are chemically explicit states; precursor-product edges preserve transformation context.

B

Node quantum chemistry and edge-resolved flux

GFN2-xTB
+ ALPB(water)

 $\Delta\omega$

electrophilicity

 ΔIP

ionization

 ΔEA

electron affinity

 $\Delta\text{G}_{\text{solv}} / \text{local}$

other edge responses

For each edge $p \rightarrow m$, transformation features are $\Delta X = X_m - X_p$.

C

Locked representation families

ECFP

2048-bit
parent structure

StateCore

8 state descriptors

FluxCore

16 edge/path descriptors

StateFluxCore

24 descriptors

The comparison asks whether state/flux chemistry adds information beyond a strong structural baseline.

D

Leakage-resistant predictive validation + higher-level quantum validation

Scaffold CV

DILrank 2.0
n = 376
10 × 5 folds

Temporal

strict n = 113
novel n = 94

DILmap

strict n = 34
novel n = 25

Cross-endpoint

genotoxicity
n = 5,367

DFT check

24 edges
144 Q-Chem jobs

Predictive value and quantum-chemical fidelity are evaluated as separate claims.